NAMAA Community at HalluScoring-2026: Beyond Predictive Performance: Reliability and Generalization in Arabic Hallucination Detection

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Abstract

Reliable Arabic hallucination detection requires not only strong ranking performance but also robustness to distribution shift, calibrated probabilities, and useful uncertainty for selective prediction. We present the NAMAA Community's submission to HalluScoring 2026 Task 1.1 and a set of post-submission diagnostic experiments. The scored, template-compliant system is a CAMeLBERT cross-encoder over the question and generated answer; it achieves 0.7596 ROC-AUC on the hidden test set, ranking third among seven teams. We compare this template-defined input path with reference-aware answer-pair models that use the gold answer. A CAMeLBERT input-framing ablation raises the unseen-question ROC-AUC from 0.8713 to 0.9272, while similar gains appear across ARBERT and MARBERTv2. Reference-free distillation recovers only a small portion of this advantage, whereas LoRA-InfoNCE substantially improves direct cosine discrimination. On a separately trained CAMeLBERT reliability replicate, temperature scaling reduces expected calibration error from 0.0993 to 0.0562, and MC-Dropout variance reduces retained-set error from 14.08% to 8.35% when the 30% most uncertain cases are deferred. These results motivate reporting distributional generalization, calibration, and selective risk alongside ROC-AUC. Code and checkpoints for the scored submission are publicly available. [1]

1 Introduction

Arabic hallucination detectors are commonly compared by ROC-AUC, but deployment also depends on generalization to unseen questions, calibrated probabilities, and the ability to defer uncertain cases. Large language models can produce fluent Arabic answers that are unsupported or factually incorrect, while dedicated Arabic hallucination resources remain limited relative to English. Recent benchmarks and shared tasks have begun to narrow this gap (Ji et al., 2023; Alansari and Luqman, 2026; Mubarak et al., 2025).

HalluScoring 2026 Task 1.1 evaluates hallucination detection on questions unseen during training(Alansari and Luqman, 2026). Each released instance contains a question, a gold reference answer, and an LLM-generated answer. However, the protected inference cell in the organizer-provided submission template encodes only the question and generated answer. Our scored, template-compliant CAMeLBERT system follows this template-defined, reference-free path. We separately investigate reference-aware answer-pair models using the gold answer but treat them exclusively as development-set diagnostics.

A single leaderboard ROC-AUC does not capture all relevant behavior. Performance may change when questions or generator/topic distributions differ from those observed during training; probabilities may be poorly calibrated; and uncertainty estimates may or may not identify cases that should be deferred. We therefore use the shared task as the basis for a broader analysis and our contributions are threefold:

- We report a template-compliant CAMeLBERT cross-encoder that achieves a 0.7596 hidden-test ROC-AUC, ranking third among seven teams, and distinguish this scored result from all development-only follow-up analyses.
- We compare template-defined question-answer framing with reference-aware answer-pair framing on unseen questions and held-out generator/topic evaluations and examine reference-free transfer through

[1]https://github.com/NAMAA-ORG/HalluScoring-Arabic-Hallucination-Detection

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knowledge distillation and LoRA-InfoNCE representation learning.

- We study calibration and selective prediction using temperature scaling, MC-Dropout, risk-coverage analysis, and model fusion and characterize the high-confidence errors that remain after calibration.

2 Related Work

Hallucination detection has been studied through response-level factuality benchmarks, supervised classifiers, retrieval-based verification, entailment-based consistency models, and LLM-based judges (Li et al., 2023; Ji et al., 2023). Although much of this research has focused on English, recent work has begun to develop resources for Arabic hallucination evaluation. HalluScore provides the Arabic question-answering benchmark underlying HalluScoring-2026, with verified reference answers and human-annotated hallucination labels (Alansari and Luqman, 2026). IslamicEval 2025 further introduced hallucination detection and correction tasks for sensitive Arabic religious content (Mubarak et al., 2025).

Our encoder-based experiments build on Arabic pretrained language models, including ARBERT and MARBERT (Abdul-Mageed et al., 2021), AraBERT (Antoun et al., 2020), and CAMeL-BERT (Inoue et al., 2021). These models provide strong Arabic representations, but their robustness under changes in questions, generator families, and available reference information remains an important empirical question.

Predictive reliability is distinct from ranking performance. MC Dropout estimates predictive variation through repeated stochastic forward passes (Gal and Ghahramani, 2016), while temperature scaling provides post-hoc probability calibration (Guo et al., 2017). Selective classification evaluates the risk-coverage trade-off obtained when uncertain predictions are deferred (Geifman and El-Yaniv, 2017). For reference-free transfer, we examine knowledge distillation (Hinton et al., 2015) and parameter-efficient representation learning through LoRA (Hu et al., 2022) combined with an InfoNCE-style contrastive objective (Gao et al., 2021).

3 Task & Experimental Setup

The released training and development files contain 4,705 and 1,300 instances, respectively; the development set covers 900 unique questions. The hidden test contains 1,030 instances over 206 questions. Training/development rows include a question Q, the gold answer A_{gold}, a generated answer A_{gen}, and a binary gold annotation $y \in \{0, 1\}$. This annotation provides supervision only: systems submit a continuous hallucination score $s = p(y = 1 \mid Q, A_{gen}) \in [0, 1]$. The official inference path is reference-free.

Track	Inference input	Evaluation role
Official CAMeL-BERT	$Q + A_{gen}$	Hidden-test submission
Reference-aware	$A_{gold} + A_{gen}$	Reference-access diagnostic
Distillation	$Q + A_{gen}$	Teacher–student transfer
LoRA–InfoNCE	$Q + A_{gen}$	Gold-free representation test
Uncertainty	$Q + A_{gen}$	Calibration and selective risk

Table 1: Experimental tracks are not a single common leaderboard

$$x_{QA} = [Q; A_{gen}], \tag{1}$$

whereas our diagnostic consistency experiments use

$$x_{NLI} = [A_{gold}; A_{gen}]. \tag{2}$$

Because x_{NLI} contains privileged information, NLI results are never reported as alternative official scores.

For internal comparisons, we use full development AUC, **Clean AUC** on questions not observed in training, and **reference-aware held-out generator** on answers produced by held-out generator families where available. Results from different tracks are interpreted within their own protocols: the official system, reference-aware experiments, representation-learning experiments, and uncertainty studies do not all share the same inference information or evaluation subset.

We additionally report ROC-AUC on a held-out generator/topic split containing $n = 200$ instances from 2 held-out generators. Because generator identity and topic are structurally coupled in this split, this evaluation measures joint generator/topic shift rather than an isolated generator effect. The split is grouped by question to prevent question overlap across partitions.

4 System Overview

Official cross-encoder. The officially scored submission is a reference-free sequence-classification pipeline: the question and generated answer are paired as $[Q; A_{gen}]$, encoded with CAMeL-Lab/bert-base-arabic-camelbert-mix, and mapped to the hallucinated-class softmax probability used as the submitted continuous score. No prompted generative LLM or gold answer is used on the official inference path. We apply no additional Arabic-specific normalization before tokenization; the released question and generated-answer strings are passed directly to the pretrained model tokenizer. Training uses AdamW, learning rate 2×10^{-5}, batch size 16, five epochs, warmup ratio 0.1, weight decay 0.01, maximum sequence length 512, and seed 42.

Reference-aware models. We recast detection as consistency between A_{gold} and A_{gen} and fine-tune CAMeLBERT, ARBERT, MARBERTv2, and multilingual variants. Soft-vote ensembles average hallucination probabilities. The strongest internal NLI ensemble reaches 0.9519 clean AUC, but it was not evaluated on the final hidden test. This ensemble includes sawb-arabert-large, a checkpoint previously trained on 2,904 external Arabic hallucination examples; the officially scored CAMeLBERT submission does not use this resource. Appendix B.1 reports the full ensemble progression.

Reference-free transfer and representation learning. A MARBERTv2 student is trained from an NLI teacher with

$$\mathcal{L}_{KD} = (1-\alpha)\mathcal{L}_{CE} + \alpha T^2 D_{KL}(p_t^T \| p_s^T), \quad (3)$$

where $T = 2$ and $\alpha = 0.3$ (Hinton et al., 2015). A separate LoRA-InfoNCE experiment freezes CamelBERT-MSA, adds LoRA adapters ($r = 16$, $\alpha = 32$) and a $768 \to 768 \to 128$ projection head, and trains in-batch contrastive pairs. The task-specific component contains 1,278,848 trainable parameters (about 4.89 MB).

Uncertainty. The single-model reliability study uses a separately trained CAMeLBERT replicate with the same reference-free architecture and $Q + A_{\text{gen}}$ input framing as the scored submission. This replica is distinct from the official checkpoint. We perform 30 MC dropout passes, apply temperature scaling, and evaluate uncertainty based on selective prediction. A separate dynamic-fusion study combines CAMeLBERT, AraBERT, and XLM-R using per-instance MC-Dropout uncertainty. Fusion hyperparameters are selected from out-of-fold training predictions, with the development set reserved for final evaluation.

5 Results & Analysis

5.1 Reference access and generalization

Table 2 compares the reference-free and reference-aware variants across evaluation settings. Only the scored CAMeLBERT system was evaluated on the hidden test (n=1,030), achieving 0.7596 ROC-AUC; E01–E06 are development diagnostics. Using the gold answer consistently improves performance for both ARBERT and MARBERTv2.

System	Full dev ($n = 1{,}300$)	Clean dev ($n = 800$)	Held-out ($n = 200$)
Reference-free ($Q + A_{gen}$)			
CAMeLBERT (scored)	.9263	.8713	–
CAMeLBERT QA (E03)	.9015	.8618	.7513
ARBERT QA (E02)	.9084	.8732	.7889
MARBERTv2 QA (E01)	.9016	.8751	.7589
MARBERTv2 KD (E06)	.9004	.8799	.7637
Reference-aware ($A_{gold} + A_{gen}$)			
ARBERT NLI (E04)	.9651	.9355	.9489
MARBERTv2 NLI (E05)	.9595	.9342	.9470
Δ ARBERT	+.0567	+.0623	+.1600
Δ MARBERTv2	+.0579	+.0591	+.1881

Table 2: ROC-AUC by evaluation split. Clean dev contains the 800 development rows whose questions do not appear in training; the held-out split contains 200 examples and measures joint generator/topic shift.

5.2 Controlled ablations, alternatives, & stability

Rather than pooling heterogeneous experiments into one ranking, Table 3 summarizes controlled within-track ablations and comparisons that answer three narrower questions: whether teacher knowledge transfers without the reference, whether contrastive training makes similarity directly useful, and whether conclusions are sensitive to fine-tuning variation. These comparisons are interpreted only within their respective protocols.

Diagnostic	Controlled comparison / result
Distillation transfer	MARBERTv2 QA → KD: +.0048 Clean AUC and +0.0048 reference-aware held-out generator.
Contrastive geometry	Cosine Dev AUC: frozen .6527; BCE 0.5318; InfoNCE **0.8746**. With RF heads: 0.9215/0.9163/0.9178
Seed stability	Reference-aware MARBERTv2: single 0.9032 Dev AUC; 3-seed probability ensemble **0.9172**.

Table 3: Targeted diagnostics for reference-free transfer, representation quality, and training stability. Values are interpreted only within each row's protocol.

Distillation recovers only a small fraction of the reference-aware advantage. InfoNCE is most useful as evidence that the embedding geometry itself becomes informative, not as a universal improvement once supervised nonlinear heads are added. The seed study separately shows that single-run development scores can vary enough to warrant stability reporting.

5.3 Calibration & Selective Prediction

The uncertainty experiments highlight the distinction between ranking performance and decision reliability. In the three-model reference-free fusion study, entropy-based dynamic weighting achieves a development ROC-AUC of 0.93236, essentially matching the best fixed weighted ensemble (0.93212) and the logistic-regression meta-learner (0.93184). The fixed weights (CAMeLBERT/AraBERT/XLM-R = 0.40/0.25/0.35) are selected from out-of-fold training predictions rather than the development set. Thus, uncertainty provides little additional ranking benefit in this setting. Table 9 in Appendix B.3 reports the corresponding bootstrap intervals.

The benefit of uncertainty is more apparent for the separately trained CAMeLBERT reliability replicate. On the 1,300-instance development set, the replicate achieves 0.9155 ROC-AUC and 0.8592 accuracy, compared with 0.9263 ROC-AUC for the official CAMeLBERT system. Table 4 summarizes the resulting reliability metrics. Temperature scaling improves probability calibration, while MC-Dropout variance achieves the lowest area under the risk-coverage curve among the evaluated selection signals. Appendix C presents the complete calibration and risk-coverage analyses.

Reliability measure	Before	After / selected
ECE ↓	0.0993	**0.0562**
NLL ↓	0.5401	**0.4027**
Brier ↓	0.1227	**0.1148**
Retained-set error ↓	14.08% (100% cov.)	**8.35%** (70% cov.)

Table 4: Core reliability results for the separately trained reference-free CAMeLBERT reliability replicate on development data.

6 Error Analysis

At threshold 0.5, the CAMeLBERT reliability replicate correctly classifies 1,117 of 1,300 development examples (accuracy 0.8592; Macro-F1 0.8568), with 473 TN, 644 TP, 74 FP, and 109 FN (Appendix C). The larger false-negative count shows that some hallucinated answers remain undetected.

Calibration does not eliminate confidently wrong predictions: among 987 predictions with calibrated confidence $\geq$ 0.90, 91 are wrong (9.22%): 64 FN and 27 FP. The gold answer is used only for post-hoc inspection and is never provided to CAMeLBERT at inference. These examples show why uncertainty supports triage but cannot replace factual verification of individual cases.

Error		Arabic example
False negative	**Q:**	من مؤسس علم التاريخ عند العرب؟
	Generated:	ابن هشام
	Gold:	اليعقوبي
		$p(\text{hall}) = .0154$; conf. $= .9846$
False positive	**Q:**	ما أكثر عنصر في القشرة الأرضية؟
	Generated:	الأكسجين
	Gold:	الأكسجين
		$p(\text{hall}) = .9480$; conf. $= .9480$

Table 5: Two verified high-confidence errors made by the CAMeLBERT reliability replicate.

7 Limitations

Several limitations should be considered when interpreting these results. Most analyses are evaluated only on the development data, limiting conclusions about generalization to the hidden test set. Reference-aware models require gold answers unavailable in the official setting, and calibration is evaluated in-sample.

8 Conclusion

In this study, we provide a broader study of generalization and predictive reliability for

Arabic hallucination detection. Our template-compliant reference-free CAMeLBERT cross-encoder achieved 0.7596 ROC-AUC on the hidden test set, ranking third among seven teams. Development-set analyses showed that reference-aware answer-pair framing substantially improves discrimination, while uncertainty estimation improves calibration and selective prediction without increasing ROC-AUC. Future work should evaluate these methods under independent calibration data and more controlled distribution shifts.

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Appendix

A Reproducibility and Implementation Details

This section collects the input templates and core settings needed to reproduce the main comparisons. The official submission is an encoder classifier rather than a prompted generative LLM, so there is no generation prompt or decoding configuration for the official inference path.

For the clean-development diagnostic used in the official-system experiment log, 800 development rows are retained because their questions do not occur among the 2,221 unique training questions; roughly 500 of the 1,300 development rows correspond to overlapping questions. This split is used as a generalization diagnostic rather than as an alternative official score. For the official CAMeLBERT run, the released Arabic strings are not subjected to an additional normalization stage. The question and generated answer are tokenized as a paired input with a maximum length of 512.

Setting	Input template
Official / reference-free	[CLS] question [SEP] model_answer [SEP] [CLS] gold_answer [SEP]
Reference-aware diagnostic	model_answer [SEP]

Table 6: Cross-encoder input templates. The official submission never uses the gold answer at inference.

Experiment	Key configuration
Official CAMeLBERT	$Q + A_{gen}$; max 512; 5 epochs; batch 16; seed 42.
ARBERT QA (E02)	$Q + A_{gen}$; max 256; 2 epochs; batch 16; seed 42.
ARBERT NLI (E04)	$A_{gold} + A_{gen}$; max 256; 4 epochs; batch 16; seed 42.
MARBERTv2 QA (E01)	$Q + A_{gen}$; max 256; 2 epochs; batch 16; seed 42.
MARBERTv2 NLI (E05)	$A_{gold} + A_{gen}$; max 256; 3 epochs; batch 16; seed 42.
MARBERTv2 KD (E06)	$Q + A_{gen}$; max 256; 2 epochs; $T = 2$, $\alpha = .3$.
CAMeLBERT uncertainty	$Q + A_{gen}$; max 512; 5 epochs; batch 16; seed 42; 30 MC-Dropout passes.

Table 7: Core training configurations. E01–E06 use learning rate 2×10^{-5}, weight decay .01, and 10% warmup.

LoRA-InfoNCE freezes CamelBERT-MSA and trains LoRA adapters ($r = 16$, scaling 32) plus a $768 \rightarrow 768 \rightarrow 128$ projection head, for 1,278,848 trainable task-specific parameters. The CAMeLBERT uncertainty study uses the same reference-free model and input framing as the official architecture, 30 MC-Dropout passes, 15 ECE bins, a 0.90 high-confidence threshold, a 0.5 classification threshold, and a temperature of 1.6203 fitted on the same 1,300-example development set used for calibration evaluation.

B Additional Diagnostic Results

These supporting results are grouped by protocol and should not be read as a common leaderboard.

B.1 Reference-aware ensemble progression

Variant	Models	Clean
S23	CAMeLBERT + MARBERTv2	.9373
S24v	S23 + ARBERT	.9478
S25v	S24v + mBERT	.9497
S25v2	CAMeLBERT + ARBERT + mBERT + sawb-arabert-large	**.9519**

Table 8: Reference-aware ensemble progression. Values are diagnostic Clean AUC, not hidden-test scores.

S23 was scored on the CodaBench development leaderboard; later variants were internal only. S25v2 includes sawb-arabert-large, previously trained on 2,904 labeled examples from a prior Arabic hallucination project. This external resource is not part of the official CAMeLBERT submission.

B.2 Additional official-system ablations and negative results

Holding CAMeLBERT fixed while changing the input framing from reference-free QA (S01) to reference-aware NLI (S02) raises Clean AUC from 0.8713 to 0.9272 (+0.0559), reinforcing the effect of direct reference access. Two discarded runs provide additional negative evidence: a zero-shot LaBSE cosine-similarity baseline (S22) reaches 0.6212 Clean AUC and 0.6768 full-development AUC, while focal loss on the sawb-arabert-large NLI model (S30) reaches 0.8886 Clean AUC and exhibits unstable epoch-level AUC. Neither run was used in the official submission.

B.3 Reference-free fusion

Fusion method	Dev AUC	Bootstrap 95% CI
Fixed weighted	.93212	[.91804, .94469]
Entropy dynamic	**.93236**	[.91809, .94487]
Logistic stacking	.93184	[.91749, .94465]

Table 9: Three-model reference-free fusion. Fixed weights were selected on out-of-fold training predictions (CAMeLBERT/AraBERT/XLM-R = 0.40/0.25/0.35).

C Reliability and Error Analysis

The reliability analysis uses the separately trained reference-free CAMeLBERT reliability replicate,

distinct from the official checkpoint. The figures and tables below follow the same sequence as the main discussion: calibration, selective prediction, then threshold-level classification errors.

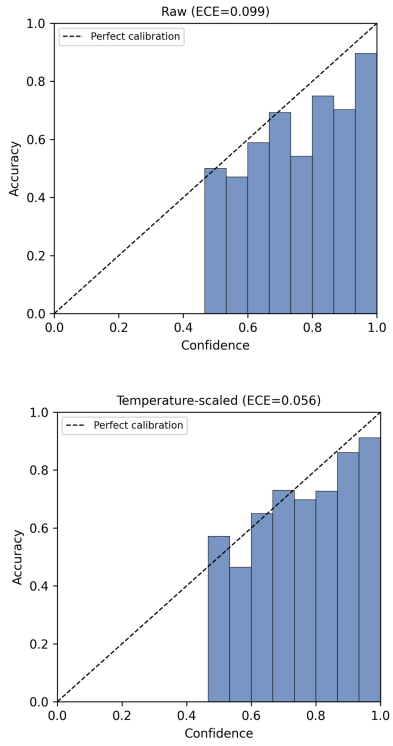


Figure 1: CAMeLBERT reliability diagrams before and after temperature scaling.

Estimate	ECE	NLL	Brier
Raw	.0993	.5401	.1227
Temp.-scaled	**.0562**	**.4027**	**.1148**
MC-Dropout mean	.0829	.4659	.1171

Table 10: CAMeLBERT calibration diagnostics; lower is better.

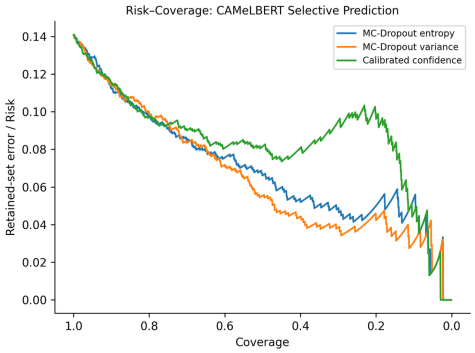


Figure 2: Selective risk–coverage for CAMeLBERT. Lower retained-set risk is better as uncertain cases are deferred.

Signal	AURC	R@90	R@70	R@50
MC-Dropout variance	**.0655**	.1145	**.0835**	**.0554**
MC-Dropout entropy	.0704	**.1128**	.0857	.0662
Calibrated confidence	.0852	.1154	.0901	.0815

Table 11: CAMeLBERT selective-prediction diagnostics. R@k is retained-set error at k% coverage; lower is better.

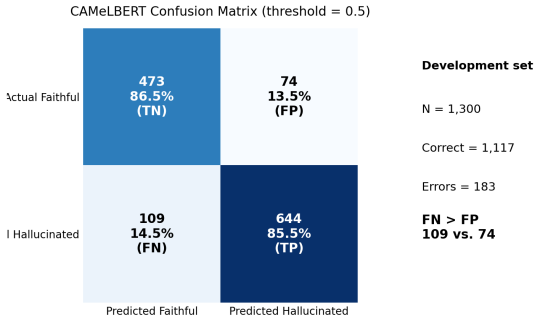


Figure 3: CAMeLBERT development confusion matrix at threshold 0.5: 473 TN, 74 FP, 109 FN, and 644 TP. False negatives are more frequent than false positives at this operating point.

D Dataset Structure

The task uses a binary gold annotation, while systems output a continuous hallucination score in $[0, 1]$. Topic tags in the dataset-structure figure below are inferred from training identifiers and are not official annotations. Topic and generator are structurally paired in the released split, so the displayed rates are descriptive rather than independent topic or generator effects.

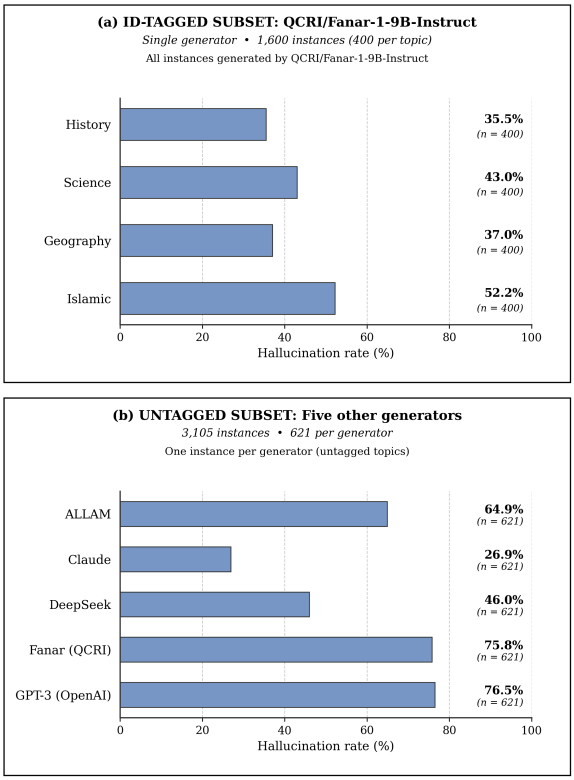


Figure 4: Training-set structure and observed hallucination rates. Top: the ID-tagged subset contains four inferred topic tags generated by QCRI/Fanar-1-9B-Instruct. Bottom: the untagged subset contains 621 instances from each of five other generators.